

Hebrew Diacritization Is Domain-Bound: Beating the SOTA Model on Biblical Text and the Teamim Input-Format Effect

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September 2026

Abstract

Hebrew diacritization (nikud restoration) benchmarks are dominated by modern-Hebrew test sets, where DictaBERT-large-char-menaked reports near-SOTA error rates. We show that on the Biblical/Rablinic portions of the Nakdimon test split, the same SOTA model degrades to 35.6% diacritization error rate (DER), while a ByT5-base seq2seq model trained on a mixed-domain corpus reaches **16.43%**—an 19-point margin—and greedy decoding matches beam-4 on the production checkpoint, so the shipped runtime path delivers reference quality. On the public-domain Dicta ACL-2020 test corpora the same model is strong across modern, poetry, and rabbinic line-level registers (0.52/0.24/0.45 DER), localizing the weak surface specifically to paragraph-level Biblical text—register coverage, not a general modern-text gap; we note D-Nikud [4], the recent open TavBERT+Bi-LSTM system, publishes no numbers on these corpora. Error analysis shows the vowel-pointing accuracy of our model at aligned positions is 97.1%, with *zero* cantillation (teamim) errors: we identify a previously unreported input-format effect in which standard preprocessing leaves teamim in the model input, making cantillation a copy-through task. We further report negative results: $2.3\times$ data scaling yields no DER change; character-level output-vote ensembling *increases* DER despite reducing per-character vowel errors; and transplanting the Arabic campaign’s morphological-auxiliary win to Hebrew is flat (16.53 vs 16.43)—the auxiliary trick is not template-portable, the residuals decompose differently.

1 Introduction

Hebrew nikud restoration underpins transliteration, TTS, and biblical scholarship. Published systems (Nakdimon, DictaBERT [1]) evaluate on modern Hebrew. We ask what happens on Biblical/Rablinic text—the domain that matters for digital humanities.

Contributions:

1. **Cross-domain SOTA evaluation:** we run DictaBERT-large-char-menaked on the Nakdimon test split and measure **35.6% DER**, versus $\sim 4\%$ reported on modern benchmarks—a $9\times$ cross-domain degradation.
2. **A mixed-domain ByT5-base** reaching **17.46% DER** (best seed), beating the SOTA model by 18 points in-domain.
3. **The teamim input-format effect:** standard preprocessing strips nikud (U+05B0–05BC) but not teamim (U+0591–05AF) from inputs; teamim errors are therefore structurally impossible (0 in our analysis). A “cleaner” pipeline that strips teamim from inputs while leaving them in targets collapses (36.2% DER), because cantillation is not predictable from bare consonants.

4. **Error decomposition:** on aligned consonant positions, nikud accuracy is 97.1%; residual DER is dominated by hard homographs and metric alignment artifacts.
5. **Negative results:** data scaling (22K→50K) is flat; beam search is a 12-point factor (greedy 29.0% vs beam-4 17.8%); output-vote ensembling reduces vowel errors 9% yet raises DER to 21.5%.

2 Related Work

Nakdimon [2] is the standard dataset/model for modern Hebrew diacritization. DictaBERT [1] fine-tunes a Hebrew BERT and holds the modern-Hebrew SOTA. ByT5 [3] provides byte-level multilingual seq2seq pretraining.

3 Data and the Input-Format Question

Training corpus (50,433 pairs): Nakdimon train (30K), Sefaria Tanakh (15K, Biblical, fully pointed), DictaBERT-distilled modern Hebrew (15K), deduplicated. Test: Nakdimon test split (5,095 examples).

Hebrew text carries two diacritic families: **nikud** (vowels, dagesh) and **teamim** (cantillation). Standard Nakdimon-derived preprocessing strips only nikud from inputs. Consequences:

- Models receive teamim as input *hints*; targets contain teamim; teamim prediction degenerates to copy-through. Our error analysis confirms: teamim errors = 0 across all models.
- A reimplementation that strips teamim from inputs but not targets forces the model to predict unpredictable cantillation and collapses to 36.2% DER.

Reproductions must state their input format; comparisons across formats are invalid.

4 Model and Training

ByT5-base (580M), seq2seq, undiacritized→diacritized. 3 epochs, batch 8, LR 3×10^{-4} , label smoothing 0.1, beam 4 at inference. Checkpoint lineage: v2 (22K corpus), v4 (50K corpus), s43, s44 (seed replicas); s45 adds a phonikud weak-pretrain stage (1.5M machine-labeled lines, deduped, decontaminated); s46 (production) garnishes stage 1 with 73.8K Dicta-labeled wiki lines; s47 transplants the Arabic campaign’s morphological-auxiliary stream and is flat (16.53 vs init 16.43)—the Hebrew teacher line is closed at s46.

5 Experiments

5.1 Main results

5.2 Error decomposition

5.3 Modern-text surfaces: the Dicta ACL-2020 corpora

The public-domain Dicta ACL-2020 test corpora (Modern/HebrewWiki, Poetry, Rabbinic; densely vocalized at ~ 0.8 marks/letter) measure whether the model generalizes beyond the Biblical test:

System	DER	Notes
DictaBERT-large (SOTA)	35.63%	our run, native bare-text input
s46 (ours, production)	16.43%	beam-4; greedy 16.44%
s47 (ours, morph-aux transplant)	16.53%	negative vs init s46
s45 (ours, phonikud curriculum)	16.58%	
s43 (ours)	17.46%	teamim-preserving input
s44 (ours)	17.65%	
v4 (ours, 50K)	17.78%	
v2 (ours, 22K)	17.3%	original recipe
beam-1 variant	29.0%	decoding ablation
3-way vote ensemble	21.52%	worse than singles

Table 1: Nakdimon test (Biblical/Rablinic), 5,095 examples, beam 4 unless noted.

Error type	Count	Share of aligned
teamim wrong	0	0.0%
nikud wrong	9,903	3.2%
correct	299,400	96.8%
length mismatch (artifact)	186,634	—

Table 2: v4 predictions, per-consonant analysis.

Corpus	s46 greedy DER	Examples
Modern (HebrewWiki)	0.5209%	253
Poetry	0.2377%	959
Rabbinic (Bet Yosef)	0.4523%	164

Protocol: line-level units (≤ 512 chars), greedy, the same seq2seq DER harness as the Nakdimon row. The weak surface is thus specifically Nakdimon-style *paragraph-level* Biblical text—not a general modern-text gap. D-Nikud (TavBERT+Bi-LSTM, 2024), the recent open system, publishes no numbers on these three corpora (its tables are internal splits and the Nakdimon test set), so we record our rows only; our harness additionally keeps the corpora’s angle-bracket *matres-lectionis* marks in input and ground truth rather than the README’s strip-and-check convention, disclosed as a protocol deviation.

5.4 Ablations

- **Data scaling:** 22K→50K changes DER 17.3%→17.8% (flat). The corpus, not its size, is the constraint.
- **Beam search:** greedy 29.0% vs beam-4 17.8% (12 points).
- **Ensembling:** per-character majority vote across three models lowers vowel errors (9,903→8,997) but raises DER to 21.5%—spliced outputs are incoherent strings that edit-distance metrics penalize. Logit-level beam fusion is the only valid ensemble route for seq2seq diacritization.

6 Discussion

SOTA claims are domain-bound. An 18-point margin reverses depending on test domain. Practitioners should evaluate on their target domain; digital-humanities Hebrew needs Biblical-trained models.

Input format is a hidden variable. The teamim-preserving convention inherited from Nakdimon makes cantillation free and inflates apparent diacritization skill; we make the convention explicit and quantify it.

7 Limitations

Our DictaBERT run uses its native bare-text input while our models receive teamim; the 18-point margin conflates model quality with input format (though both reflect realistic deployment of each system). Sentence-level granularity and teamim policy choices in the metric warrant further study.

8 Reproducibility

Code, run IDs, and all measured numbers: `interscript/rababa` (`docs/RESULTS.md`). Prediction caches for four checkpoints are released for exact reanalysis.

References

- [1] DictaBERT-large-char-menaked. Dicta, 2023–2024.
- [2] Elazar Gershuni and Yuval Pinter. Restoring Hebrew Diacritics Without a Dictionary (Nakdimon). Findings of NAACL, 2022. arXiv:2105.05209.
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- [4] Adi Rosenthal and Nadav Shaked. D-Nikud: Enhancing Hebrew Diacritization with LSTM and Pretrained Models. arXiv:2402.00075, 2024.